7.c.c Gait with an Instantaneous Centre of Curvature

The ICC (Instantaneous Centre of Curvature) is a point in the X, Y plane relative to the scorpion's origin that the scorpion will turn about. The ICC has X, Y coordinates and a direction of rotation (Either Clock Wise or Counter Clock Wise).

With this information the scorpion is able to calculate any trajectory and walk it in it. This method allows seamless transitions from walking or turning to walking or turning in any other direction.

Figure 9 shows a representation of the ICC relative to the scorpion's origin. The scorpion's forward direction is positive X and subsequently left is positive Y is the scorpions left. The green circles represent the base position of the legs, the red circle is the ICC and the blue circle is the origin of the scorpion. The dotted circle segments are the paths that each respective leg will follow either in a clockwise or counter clockwise direction dependant on what the ICC was set as. The scorpion's origin will move at a tangent to the dash dot line. The stride length of each leg is scaled proportionally to the distance it is from the ICC which ensures that the stride length is never longer than the diameter of the circle as that would cause the target position to be incorrect.

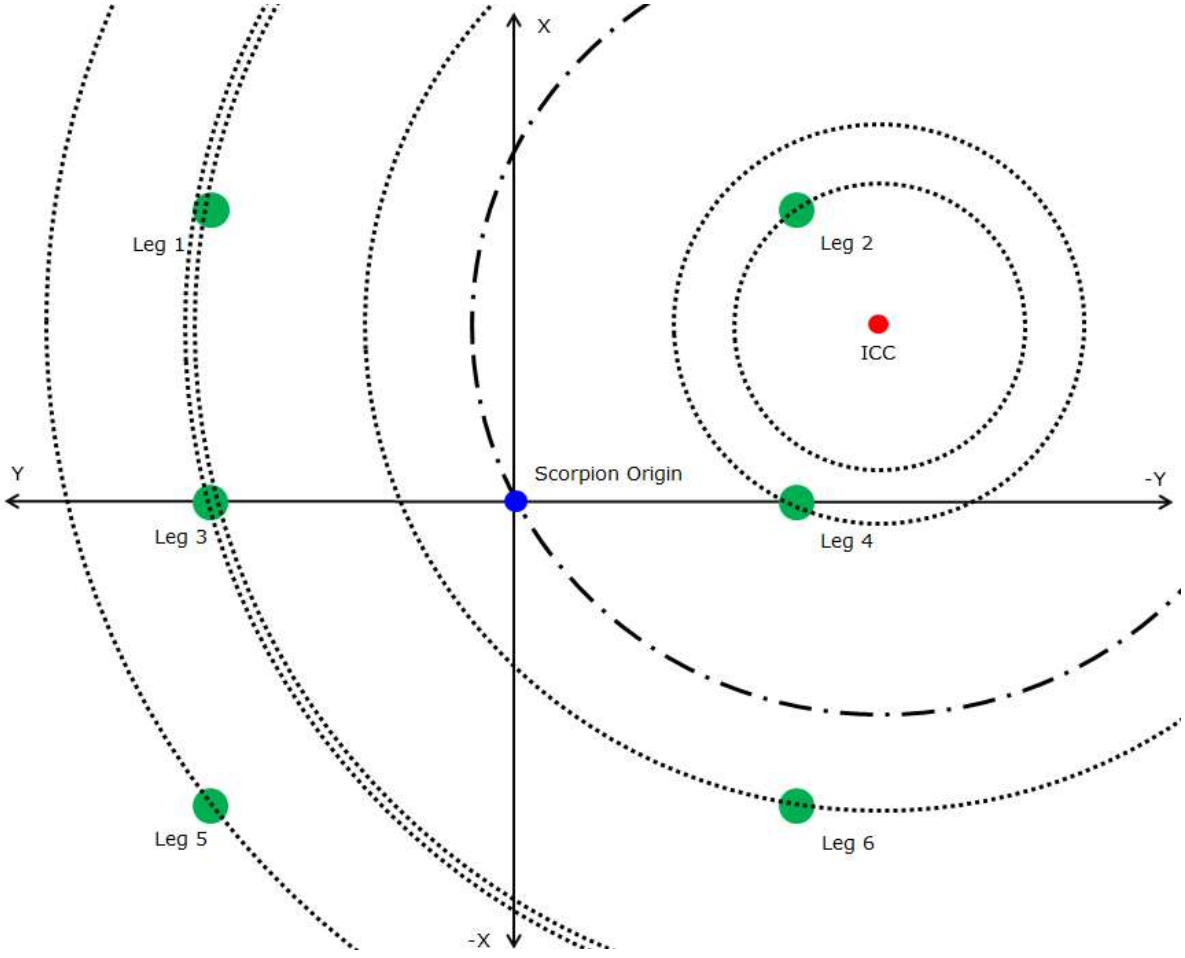


Figure 9: ICC

The equation used to scale the stride length for each leg is:

$$legxStrideLength = the\ stride\ length\ for\ the\ specified\ leg$$
$$legDistanceFromICC = the\ current\ legs\ distance\ from\ the\ ICC$$
$$maxDistanceFromICC = the\ furthest\ legs\ distance\ from\ the\ ICC$$

$$legxStrideLength = strideLength \cdot \frac{legDistanceFromICC}{maxDistanceFromICC}$$

As the leg needs to follow the circular path a number of points long the circle are calculated, the number of points is equal to the number of steps in the current phase of the leg.

$$stepsInPhase = The\ number\ of\ steps\ in\ the\ current\ phase$$
$$X_{Dist} = The\ X\ componenet\ of\ the\ distanece\ of\ the\ leg\ from\ the\ ICC$$
$$Y_{Dist} = The\ Y\ componenet\ of\ the\ distanece\ of\ the\ leg\ from\ the\ ICC$$
$$XFromICC = The\ X\ componenet\ of\ the\ target\ position\ from\ the\ ICC$$
$$YFromICC = The\ Y\ componenet\ of\ the\ target\ position\ from\ the\ ICC$$
$$\mathrm{X_{target}} = \text{The X componenet of the legs target position}$$
$$\mathrm{Y_{target}} = \text{The Y componenet of the legs target position}$$
$$ICC_X = The\ X\ coordinate\ of\ the\ ICC$$
$$ICC_Y = The\ Y4\ coordinate\ of\ the\ ICC$$

$$\alpha = 2 \cdot \sin^{-1}\left(0.5 \cdot \frac{legxStrideLength}{legDistanceFromICC}\right)$$
$$\alpha_{step} = \frac{\alpha}{stepsInPhase}$$
$$\beta = \tan^{-1}\left(\frac{Y_{Dist}}{X_{Dist}}\right)$$

Note: atan2 is used in the code to return the correct quadrant and avoid trying to divide by zero.

If the ICC's direction is clockwise:

$$\beta_{target} = \beta + \alpha_{step}$$

If the ICC's direction is counter clockwise:

$$\beta_{target} = \beta - \alpha_{step}$$
$$XFromICC = \cos(\beta_{target}) \cdot legDistanceFromICC$$
$$YFromICC = \sin(\beta_{target}) \cdot legDistanceFromICC$$
$$\mathrm{X_{target}} = XFromICC + ICC_X$$
$$\mathrm{Y_{target}} = YFromICC + ICC_Y$$

These equations are used to calculate the carget position for each leg while it is in the support or swing phase. These are recalculated for each time the step number is incremented.